



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 02 Task REPORT

IP Addressing (FLSM & VLSM) and VLAN Configuration

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Submission Date: 11 July 2026

1. Objectives

- Understand FLSM and VLSM and calculate network/broadcast addresses, subnet masks, and usable host ranges.
- Design efficient IP addressing schemes for enterprise networks.
- Understand VLAN concepts and configure/verify VLANs, access ports, and trunk links on Cisco switches.
- Implement and verify the design in both Cisco Packet Tracer and GNS3.

2. FLSM vs VLSM

FLSM applies the same subnet mask to every subnet — simple to plan but wastes addresses when host requirements differ. VLSM allows different subnet masks within the same network, sizing each subnet to its actual requirement — the standard approach in real enterprise/ISP networks.

Aspect	FLSM	VLSM
Subnet size	All equal	Sized per requirement
Efficiency	Lower (wastes addresses)	Higher (minimal waste)
Complexity	Simple	More planning required
Typical use	Labs / uniform networks	Real enterprise / ISP networks

3. FLSM Worksheet — 10 Problems

#	Base Network	Subnets Req.	Prefix	Mask	Block Size	Usable Hosts	First Subnet	Last Subnet
1	192.168.10.0/24	4	/26	255.255.255.192	64	62	192.168.10.0	192.168.10.192
2	192.168.20.0/24	8	/27	255.255.255.224	32	30	192.168.20.0	192.168.20.224
3	10.0.0.0/8	256	/16	255.255.0.0	65536	65534	10.0.0.0	10.255.0.0
4	172.16.0.0/16	16	/20	255.255.240.0	4096	4094	172.16.0.0	172.16.240.0
5	192.168.100.0/24	2	/25	255.255.255.128	128	126	192.168.100.0	192.168.100.128
6	192.168.50.0/24	16	/28	255.255.255.240	16	14	192.168.50.0	192.168.50.240
7	10.10.0.0/16	4	/18	255.255.192.0	16384	16382	10.10.0.0	10.10.192.0
8	172.20.0.0/16	32	/21	255.255.248.0	2048	2046	172.20.0.0	172.20.248.0
9	192.168.30.0/24	6 (min)	/27	255.255.255.224	32	30	192.168.30.0	192.168.30.160
10	192.168.5.0/24	32	/29	255.255.255.248	8	6	192.168.5.0	192.168.5.248

4. VLSM Worksheet — 5 Design Scenarios

Method: sort requirements largest → smallest, assign the smallest block that satisfies each one, allocate sequentially with no overlap.

Scenario 1 — Small Company LAN (Base: 192.168.1.0/24)

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
Sales	60	/26	255.255.255.192	192.168.1.0	192.168.1.1–.62	192.168.1.63
HR	28	/27	255.255.255.224	192.168.1.64	192.168.1.65–.94	192.168.1.95
IT	12	/28	255.255.255.240	192.168.1.96	192.168.1.97–.110	192.168.1.111
Management	6	/29	255.255.255.248	192.168.1.112	192.168.1.113–.118	192.168.1.119
Router Link	2	/30	255.255.255.252	192.168.1.120	192.168.1.121–.122	192.168.1.123

Scenario 2 — Campus Network, 4 Buildings (Base: 172.16.0.0/16)

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
Building A	500	/23	255.255.254.0	172.16.0.0	172.16.0.1–1.254	172.16.1.255
Building B	200	/24	255.255.255.0	172.16.2.0	172.16.2.1–.254	172.16.2.255
Building C	100	/25	255.255.255.128	172.16.3.0	172.16.3.1–.126	172.16.3.127

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
Building D	50	/26	255.255.255.192	172.16.3.128	172.16.3.129–.190	172.16.3.191
WAN Link	2	/30	255.255.255.252	172.16.3.192	172.16.3.193–.194	172.16.3.195

Scenario 3 — Small Branch Office (Base: 10.0.0.0/24)

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
Finance	40	/26	255.255.255.192	10.0.0.0	10.0.0.1–.62	10.0.0.63
Marketing	20	/27	255.255.255.224	10.0.0.64	10.0.0.65–.94	10.0.0.95
Reception	8	/28	255.255.255.240	10.0.0.96	10.0.0.97–.110	10.0.0.111
Server Room	4	/29	255.255.255.248	10.0.0.112	10.0.0.113–.118	10.0.0.119
Router Link	2	/30	255.255.255.252	10.0.0.120	10.0.0.121–.122	10.0.0.123

Scenario 4 — ISP Customer Block Allocation (Base: 192.168.50.0/24)

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
Customer A	100	/25	255.255.255.128	192.168.50.0	192.168.50.1–.126	192.168.50.127
Customer B	50	/26	255.255.255.192	192.168.50.128	192.168.50.129–.190	192.168.50.191
Customer C	25	/27	255.255.255.224	192.168.50.192	192.168.50.193–.222	192.168.50.223
Customer D	10	/28	255.255.255.240	192.168.50.224	192.168.50.225–.238	192.168.50.239
Uplink	2	/30	255.255.255.252	192.168.50.240	192.168.50.241–.242	192.168.50.243

Scenario 5 — VLAN Lab Addressing Plan (Base: 172.20.10.0/24)

Segment	Hosts	Prefix	Mask	Network	Usable Range	Broadcast
VLAN 10 – Sales	50	/26	255.255.255.192	172.20.10.0	172.20.10.1–.62	172.20.10.63
VLAN 20 – HR	25	/27	255.255.255.224	172.20.10.64	172.20.10.65–.94	172.20.10.95
VLAN 30 – IT	10	/28	255.255.255.240	172.20.10.96	172.20.10.97–.110	172.20.10.111
Router Link	2	/30	255.255.255.252	172.20.10.112	172.20.10.113–.114	172.20.10.115

5. VLAN Lab Design (Packet Tracer)

Addressing plan from VLSM Scenario 5 (172.20.10.0/24) used to build a router-on-a-stick topology: Router1, two 2960 switches (Switch0, Switch1), and 6 PCs across 3 VLANs.

5.1 VLAN Plan

VLAN	Name	Subnet	Gateway	Switch Ports
10	Sales	172.20.10.0/26	172.20.10.1	Switch0 Fa0/1, Fa0/2
20	HR	172.20.10.64/27	172.20.10.65	Switch0 Fa0/3, Switch1 Fa0/1
30	IT	172.20.10.96/28	172.20.10.97	Switch1 Fa0/2, Fa0/3

5.2 Trunk Links

Link	Port A	Port B	Mode
Switch0 ↔ Router1	Switch0 Gi0/1	Router1 Gi0/0	Trunk (802.1Q)
Switch0 ↔ Switch1	Switch0 Fa0/24	Switch1 Fa0/24	Trunk (802.1Q)

5.3 Topology

1

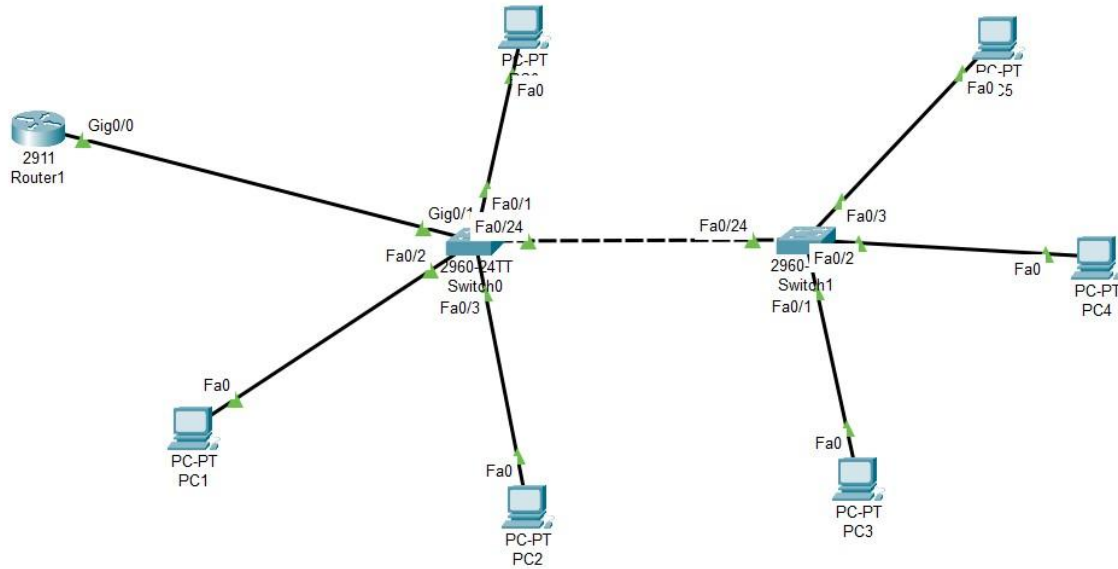


Figure: Packet Tracer topology — Router1, Switch0, Switch1, PC0–PC5.

6. Implementation — Cisco Packet Tracer

6.1 Switch0 Configuration

```
enable
configure terminal
vlan 10
  name Sales
vlan 20
  name HR
vlan 30
  name IT
exit
interface range fa0/1 -2
  switchport mode access
  switchport access vlan 10
exit
interface fa0/3
  switchport mode access
  switchport access vlan 20
exit
interface fa0/24
  switchport mode trunk
exit
interface gi0/1
  switchport mode trunk
exit
end
write memory
```

6.2 Switch1 Configuration

```
vlan 20
  name HR
vlan 30
  name IT
exit
interface fa0/1
  switchport mode access
  switchport access vlan 20
exit
```

```

interface range fa0/2 -3
  switchport mode access
  switchport access vlan 30
exit
interface fa0/24
  switchport mode trunk
exit
end
write memory

```

6.3 Router1 (Router-on-a-Stick)

```

enable
configure terminal
hostname Router1
interface gi0/0
  no shutdown
exit
interface gi0/0.10
  encapsulation dot1Q 10
  ip address 172.20.10.1 255.255.255.192
exit
interface gi0/0.20
  encapsulation dot1Q 20
  ip address 172.20.10.65 255.255.255.224
exit
interface gi0/0.30
  encapsulation dot1Q 30
  ip address 172.20.10.97 255.255.255.240
exit
end
write memory

```

PCs configured with static IP, subnet mask, and default gateway per the VLAN plan above (Section 5.1).

7. Verification

Verified with show vlan brief and show interfaces trunk on Switch0:

```

Switch>show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Gig0/2
10	Sales	active	Fa0/1, Fa0/2
20	HR	active	Fa0/3
30	IT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch>show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Fa0/24	on	802.1q	trunking	1
Gig0/1	on	802.1q	trunking	1

```

Switch>show interfaces trunk

```

Port	Vlans allowed on trunk
Fa0/24	1-1005
Gig0/1	1-1005

```

Switch>show interfaces trunk

```

Port	Vlans allowed and active in management domain
Fa0/24	1,10,20,30
Gig0/1	1,10,20,30

```

Switch>show interfaces trunk

```

Port	Vlans in spanning tree forwarding state and not pruned
Fa0/24	1,10,20,30
Gig0/1	1,10,20,30

```

Switch>

```

Figure: show vlan brief / show interfaces trunk — confirms VLANs 10 (Sales), 20 (HR), 30 (IT) and trunking on Fa0/24 & Gig0/1.

Ping tests: same-VLAN pings (e.g. PC0 ↔ PC1) succeed via switching; cross-VLAN pings (e.g. Sales ↔ IT) succeed via Router1's inter-VLAN routing.

8. GNS3 Recreation

A separate, simplified topology was built in GNS3 (task2.gns3) to isolate and verify pure Layer 2 VLAN trunking — no router involved:

- Two switches (Switch1, Switch2) connected via an 802.1Q trunk link.
- Four VPCS end devices: PC4 & PC3 on Switch1, PC1 & PC2 on Switch2.
- VLAN 10 (192.168.10.0/24) and VLAN 20 (192.168.20.0/24) configured on both switches, independent of the Packet Tracer addressing plan.
- Verified with show vlan brief and vpcs> ping between same-VLAN hosts across the trunk (e.g. PC4 ↔ PC1).

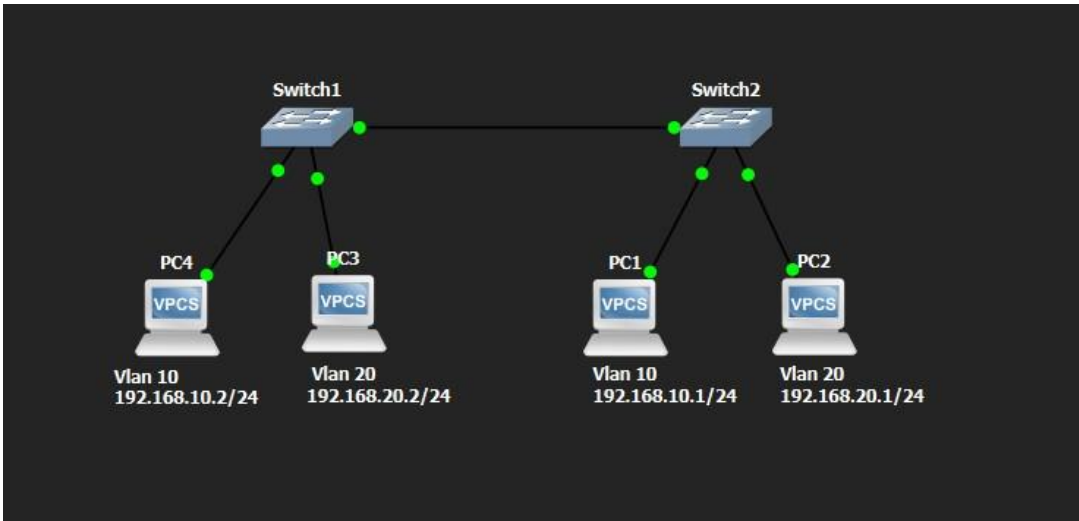


Figure: GNS3 topology — Switch1 & Switch2 with VPCS hosts PC1–PC4 across VLAN 10 and VLAN 20.

9. Challenges & Key Learnings

Challenge	Solution
Cross-VLAN ping failed initially	Verified router sub-interface dot1Q tag matched the VLAN ID and the connecting port was trunk, not access.
Same-VLAN PCs on different switches couldn't communicate	Confirmed the inter-switch link was trunked and carrying the required VLANs.
GNS3's default switch doesn't support VLAN tagging	Used VPCS end devices and switch nodes that support 802.1Q trunking.

- Explained FLSM vs VLSM and when to use each.
- Calculated subnet masks, network/broadcast addresses, and host ranges across 15 scenarios.
- Configured VLANs, access/trunk ports, and router-on-a-stick inter-VLAN routing.
- Verified configuration and connectivity using Cisco IOS commands on two platforms (Packet Tracer & GNS3).

10. Conclusion

This assignment reinforced subnetting fundamentals (FLSM and VLSM) and applied them to a realistic VLAN segmentation scenario across two simulation platforms — from address planning, through device configuration, to verification and troubleshooting, mirroring the workflow of a Network Administrator.